

Applications of Algebra (Part I): Using Systems of Equations (Homework)

In each problem below, define variables, write a system of equations to represent the problem, and then solve the system.

① Billy collects snakes and lizards. He currently has six more lizards than snakes. Altogether, he has 20 reptiles (only lizards and snakes). How many of each reptile does he have?

② Larry collects cars. He collects only Ford cars and General Motors cars. If Larry has a total of 19 cars, and he has seven fewer Ford cars than General Motors cars, how many cars of each make does he have?

③ In a park there are eucalyptus trees and pine trees only. There are four times as many pine trees as eucalyptus trees. If there are 35 trees total, how many of each tree are there?

④ Kevin picked apples and oranges from the trees in his backyard. He picked eight times as many oranges as apples and the difference between the number of oranges he picked and the number of apples he picked was 56. How many apples did he pick?

Do not use "0" as a variable. It looks like the symbol for zero. Choose a different letter.

⑤ A vendor sells peaches for \$4 each and tangerines for \$3 each. If the vendor made \$44 from peach and tangerine sales, and he sold 12 pieces of these types of fruit, how many pieces of each type of fruit did he sell?

⑥ A recipe requires teaspoons of vinegar and teaspoons of brown sugar. If the number of teaspoons of vinegar required is one less than three times the number of teaspoons of brown sugar required, and there are eleven total teaspoons of both ingredients required, how many teaspoons of brown sugar are required?

⑦ The price for admission at a hockey game is \$12 per adult and \$10 per child. On the first day that the tickets were sold online, the stadium sold 30 tickets and made \$342 in sales. How many adult tickets were sold on the first day?

⑧ A store sold nine chicken bowls and seven beef bowls for \$50. The next day the same store sold five chicken bowls and six beef bowls for \$32. Find the price for each type of bowl.

You may use a calculator for the subsequent problems.

⑨ A toy store sold four RC cars and three RC planes. The store made \$651. The next day the same store sold two RC cars and one RC plane and made \$253 in sales. Assuming that the store sells only one car model and one plane model, what are the prices per item?

⑩ Lucy bought chairs for \$31.50 each and tables for \$76.13 each. If she bought 25 of these items in total and she spent \$1,055.28, how many tables did she buy?

⑪ A blacksmith made five lamp stands and two mail posts. He charged \$642.65 for these items. Another day he made three lamp stands and two mail posts and he charged \$468.19 for these items. Assuming that the blacksmith charged the same amount for each lamp stand and each mail post (although stands and posts do not cost the same), how much did he charge for each item?

⑫ Daniel has dimes and quarters in his pocket. He knows that the total value of these coins is \$2.95 and there are 16 coins in his pocket. How many dimes are in his pocket?

⑬ 18 karat gold is 75% gold and 9 karat gold is 37.5% gold. A metallurgist has only 18 and 9 karat gold but he wants to make 20 ounces of 12 karat gold. How many ounces of each purity should he use to do this (12 karat gold is 50% gold)?

⑭ Arnold likes to mix sodas to produce his own flavors. Grape soda costs \$0.15 per ounce and orange soda costs \$0.12 per ounce. How many ounces of each soda should he use if he wants to make a ten ounce mixture that costs \$0.13 per ounce?

Answers

- ① 7 snakes & 13 lizards
- ② 13 GM's & 6 Fords
- ③ 7 eucalyptus trees & 28 pine trees
- ④ 8 apples
- ⑤ 8 peaches & 4 tangerines
- ⑥ 3 teaspoons of brown sugar
- ⑦ 21 adult tickets
- ⑧ \$4/chicken bowl & \$2/beef bowl
- ⑨ \$54/RC car & \$145/RC plane
- ⑩ 6 tables
- ⑪ \$87.23/lamp stand & \$103.25/mail post
- ⑫ 7 dimes
- ⑬ $6\frac{2}{3}$ oz 18 karat gold & $13\frac{1}{3}$ oz 9 karat gold.
- ⑭ $3\frac{1}{3}$ oz grape soda & $6\frac{2}{3}$ oz orange soda.